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The palaeomagnetism of Cambro-Ordovician redbeds, the Erquy Spilite Series and the Trégastel-Ploumanac'h granite complex, Armorican Massif (France and the Channel Islands)

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Summary. In addition to a component (A) of recent origin, two NRM components are distinguished in the Cambro-Ordovician redbeds of the Armorican Massif. In most sites other than those from northern Brittany the oldest (C) is probably Silurian or early Devonian, and is mainly carried by specularite with high blocking temperatures. This component was variably overprinted by a Devonian or early Carboniferous component (B3) which was probably acquired as a viscous PTRM on uplift after burial, and is carried by hematite pigment with intermediate to high blocking temperatures. In the red succession of Plourivo-Bréhec (northern Brittany) declination scatter of two intermediate to high blocking temperature components (B1 and B2) is consistent with clockwise rotation of the bulk of Europe during the late Carboniferous, implied independently by published European Carboniferous palaeomagnetic data.

Stable NRM in the Erquy Spilite Series yields a palaeomagnetic pole at 344° E, 35° N ($dp = 21°$, $dm = 22°$), and was probably acquired during remagnetization following Late Precambrian or early Cambrian folding. This is consistent with a middle to late Cambrian age of remagnetization estimated by comparison with other poles of known age.

A palaeomagnetic pole position at 332° E, 34° S ($dp = 4°$, $dm = 7°$) determined for the Hercynian Trégastel-Ploumanac'h complex is consistent with other middle to late Carboniferous poles from elsewhere in Europe.

1 The Armorican redbeds

1.1 INTRODUCTION

Red shales, arkoses and grits unconformably overlie deformed Brioverian, Cadomian granite or Pentevrian crystalline basement at the base of the Lower Palaeozoic synclinal basins of Normandy and Brittany. These redbeds are known from stratigraphic and faunal evidence to be Cambrian to Tremadocian in age; at Carteret in Normandy and in the Loire region of southern Brittany the age of the basal redbeds is known more precisely to be Lower Cambrian (Bigot 1925) and Tremadocian (Blaise, Cavet & Lardeux 1970) respectively.

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Table 1. Location and geology of redbeds sampled in the Armorican Massif.

Region	Geological unit and locality	Site	Lat.	Long.	Rock or formation name	Dip/dip direction	Stratigraphic relationships	Age (Myr)
Jersey	Rozel cong.;	TH	49.2°N	358°E	Red shale	Variable 20–30/000–030	Unconform. overlies 533 ± 16 Myr rhyolites	< 533 ± 16, Probably upper C
C.I.	Rozel cong.;	LS			Red shale	29/040	533 ± 16 Myr rhyolites	Probably upper C
N. Normandy	E. of Omonville-la-Rogue	UN1	49.6°N	358.4°E	Red grit	47/132	Immediately unconform. overlies Omonville gneiss	C? (Probably = Alderney sst)
	Schistes de Carteret	UN2	49.4°N	358.2°E	Red shale	37/033	Conform. overlain by shales and limest. of St. Jean-de-la-Rivière, unconform; overlain by ORD.	Lower C (tribolites; archaeocyathea)
Middle Normandy	South of Montpinchon	MN1	49°N	358.7°E	Red shale	32/345	Overlain by ORD quartzites	Probably C
	3.5 km SE of Cerisy-la-Salle	MN2			Red grit	35/018	Overlain by ORD quartzites	Probably C
S. Normandy	'Grès rouges de St Remy', 2.8 km SE of Tessy	MN3	49°N	359°E	Red sandstones	86/165	Unconform. overlies Cadomian granite (605 ± 15 Myr); conform. on lower C Poudingue pourpré.	Probably middle C
	Poudingue pourpré, Villedieu.	MN4	48.8°N	358.8°E	Coarse grit	10/358	Unconform. on Cadomian granite	Probably lower C
	Poudingue pourpré, 9 km E of Granville	MN5			Red shale ¢ clasts	25/022	Unconform. on Cadomian granite	Probably lower C
	La Haye-Pesnel	LN4	48.8°N	358.6°E	Coarse grit	20/335	Unconform. on Cadomian granite	Probably lower C

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N. Brittany Côtes-du-Nord	Cap Frehel-Erquy red succession	CD8, 9, 10, 11 and 12 and CF	48.7°N	357.6°E	Pink arkoses and sandstones	CD10: 24/331 CF: 36/248	Unconform. overlie Brioverian and Cadomian dioritic basement	Disputed: C – ORD or Devonian
	Plourivo-Bréhec red succession	CD18	48.7°N	357°E	Port Lazo and	Variable ~ 40/130	Unconform. overlies	Age disputed as for
		CD22			Bréhec red shale	10/050	Brioverian metaseds.	Frehel-Erquy redbeds,
		CD19			Red sst.	33/281		but probably equi-
		CD20			Brown sst ¢ concretions	21/238		valent to this succession
		CD21, 23			Grès rose de Toul-Lan			
		CD26	48.7°N	356.8°E	Grès rouge de la Roche-Jagu	43/080		
W. Brittany; Finistere	Cap de la Chevre, Crozon Peninsula	CP	48.2°N	355.4°E	Red shale	25/332	Unconform. overlies Brioverian metaseds; conform. overlain by Grès Armoricain (ORD)	Pre Arengian (ORD), C?
		CR			Red shale	20/340		
Loire Valley (Maine-et-Loire)	'Série du moulin de de Châteaupanne'	LV	47.4°N	359.3°E	Red grit	8/287	Unconform. overlies Brioverian metaseds.	Post C and pre, Ilanvirnian; probably Tremadocian
Central Brittany, west & south of Rennes	Poudingue de Montfort; north flank of 'Syncline, south of Rennes'	SR1	48°N (15 km SSW of Rennes)	358.2°E	Red shale (slaty)	33/193	Unconform. overlies Brioverian metaseds; conform. overlain by ORD quartzites	Pre Arenigian C?
		SR2 (Les Rochelles quarry)	48°N (10 km S of Montfort)	358.1°E	Red shale	9/073		Pre Arengian C?
		SR3 (La Marette quarry)	48°N (15 km SW of Montfort)	358°E	Red shale	36/196	Unconform. overlies Brioverian metaseds; conform. overlain by ORD quartzites	Pre Arengian C?

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Van der Voo and Bonhommet (1976) isolated similar NRM directions in redbeds near Paimpol, northern Brittany, to those in the Devonian Hyenville redbeds of Normandy. Recently, their studies of the Palaeozoic redbeds have been extended to other localities in the Armorican Massif.

1.2 SAMPLING PROCEDURE

Red shales, arkoses or grits were collected at 29 sites throughout the Amorican Massif (Table 1; Fig. 1) with the aim of improving the palaeomagnetic record of the Lower Palaeozoic of Western Europe. Fourteen of the sites were located at the base of several of the Palaeozoic basins of Brittany and Normandy, and the remaining 15 were located in the red successions of more uncertain age in Jersey and the Côtes-du-Nord (northern Brittany). Most of these were collected by *in situ* drilling. The red shale sampled at the base of the Rozel Conglomerate in north-eastern Jersey is probably middle Cambrian to middle Silurian in age (e.g. Duff 1978a). Owing to the limited thickness of the exposed basal red successions it was only possible to obtain single site collections at different localities, with drilled or hand samples typically spread over a few metres of section.

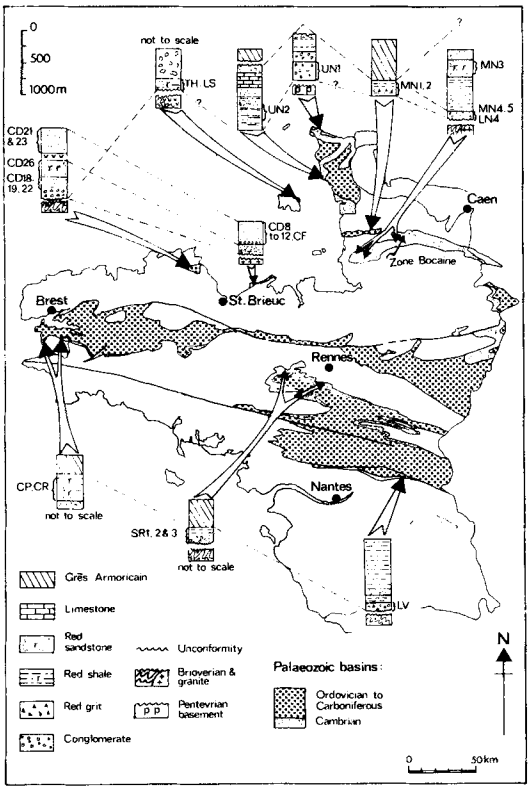


Figure 1. Palaeomagnetic sampling localities in the redbeds at the base of the Palaeozoic basins of the Armorican Massif. The nature of the Cambro-Ordovician succession at each locality and the likely correlation of these successions across the Massif are also indicated.

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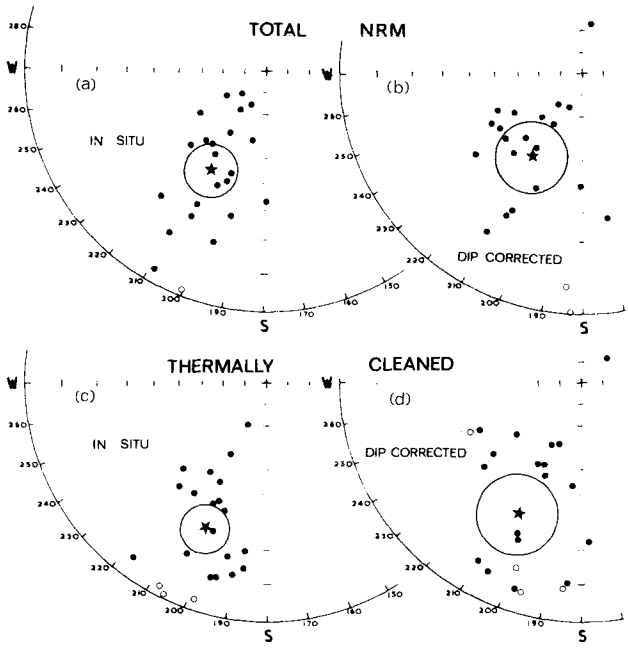


Figure 2. Site mean directions of NRM for the Armorican redbeds before and after optimum thermal cleaning. The overall mean directions are shown (stars) with their circles of 95 per cent confidence. Closed symbols: downward directions; open symbols: upward directions (stereographic projections).

1.3 DEMAGNETIZATION STUDIES OF NRM

The *in situ* and dip-corrected site mean directions of total NRM for the 29 redbed sites (Fig. 2) are downward toward the SSW with variable inclinations so that they are distributed close to a vertical plane.

One specimen from each site was progressively af demagnetized in peak fields up to 160 mT. The NRM of the red shales and grits from Normandy, Jersey and central and southern Brittany generally decayed only slightly during af demagnetization and yielded endpoints close to the direction of total NRM (Fig. 3) as expected where specular hematite or hematite pigment is the dominant magnetic phase. In contrast, the arkoses are only slightly pigmented and show changes of NRM direction and intensity in low peak fields consistent with the presence of magnetite (Fig. 3 (d) and (f)).

Acid leaching in conjunction with thermal demagnetization often yields information about the multicomponent nature of redbed NRM which is not retrievable by thermal demagnetization alone (Park 1970; Roy & Park 1974).

One or two specimens from most sites were leached in 10N HCl following the technique of Park (1970). Specimens were immersed in the acid for a given time, washed in water, measured and then replaced in fresh acid. Measurements were made over a period of eight to nine weeks and typical results are shown in Fig. 4 and Table 2. During preferential removal of the soluble hematite pigment, the resultant NRM vector rotates from the direction of the total NRM toward an endpoint corresponding to the NRM of the insoluble specular hematite and magnetite fraction. This separation of the soluble and insoluble magnetic fractions is marked by a change of specimen colour from red to drab grey or brown, and by a levelling-off in the NRM and susceptibility decay curves (Fig. 4). Changes in direction of

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Table 2. Results of acid leaching and af demagnetization of the Armorican redbeds.

	Acid leaching								Af demagnetization							
Site	Specimen	Components removed	Components left	*M*	*D*	*I*	M/M_0	Time*	Specimen	Mdf (mT)	Components removed	Components left	*M*	*D*	*I*	M/M_0
TH	1–4F	A + B3	C:	3.45	194	61	0.53	1242	1–7C	60	A	B3 + C	3.9	186	69	0.9
	1–6C	A + B3	C:	2.82	234	76	0.50	1242	1–17A	70	A	B3 + C	5.0	222	62	0.83
UN1									1–10	140	A	B3 + C	3.7	(197	46)	0.92
UN2	2–5	A	C′:	0.6	(230	68)	0.2	1344	2–4	140	A	C′	2.7	(234	63)	0.74
MN1	1–4	A	B3 + C:	1.0	(198	−11)	0.1	864	1–6	140	None	B3 + C:	9.2	(200	3)	1
					199	16								204	23	
MN2	2–6	A + B3	C:	0.4	228	67	0.1	1512	2–5	140	None	B3 + C:	3.5	(207	14)	1
														211	48	
MN3	3–1	A + C(?) + B1	B2:	1.1	(175	3)	0.5	1512	3–3	140	A	B1 + B2 + C:	3.6	(215	6)	0.97
					221	−78)								260	−35	
LN4									4–2		A, some of B1 + B2	C, some of B1 + B2				
CD10	10–6	Erratic during leaching							10–3	120	A	B3 + C:	0.4	232	51	0.84
CF									1–6	50	A, some of B1 + B2	B1 + B2:	0.4	(204	25)	0.35
														225	45	
CD18	18–1	A, some of B2	B2 + C:	7	(183	16)	1.1	144	18–13	160	A, some of B1 + B2	B1 + B2:	4.6	(201	21)	0.91
					181	−9								192	5	
CD19	19–6	Erratic during leaching							19–8	70	A, some of B1 + B2	B1 + B2	1.5	(194	29)	0.5
														210	22	
CD20	20–4	Erratic during leaching							20–6	15	A, some of B1 + B2	B1 + B2	3.4	(197	15)	0.32
														199	−1	
CD22	22–5	Some of B1 + B2	A + B1 + B2:		(216	18)			22–9	140	A	B1 + B2:	3.2	(208	15)	1.2
														208	11	
CD26	26–7	Some of B2	A + B1 + B2:		(225	6)	0.3	1272	26–1	140	A, some B2	B1 + B2:	2.7	(208	15)	0.81
														189	36	
CP	1–1	A + B3	C:	3.2	233	52	0.4	1176	1–7	160	A	B3 + C:	6.7	(198	46)	0.93
														228	58	
CR	5	A	B3 + C:	2.6	(228	33)	0.3	1176								
					242	38										
LV	1–9	A	B3 + C:	5.4	(209	42)	1.1	504	1–1	100	A	B3 + C:	5.4	(217	56)	0.98
					216	40								228	53	
SR1	1–3	A, some of B1 + B2	B1 + B2:	No endpoint					1–2	160	A + B2	B1:	2.0	212	20	0.72
SR2	2–3								2–1	160	A	B3 + C:	3.0	(208	57)	0.72
														196	62	
SR3	3–1	A, some of B1 + B2	B1 + B2	No endpoint					3–3	160	A, some B2	B1 + B2:	2.1	(197	10)	0.71
														197	−26	

Notes: *M*, *D*, *I*: Intensity, declination and inclination of the component(s) left after acid or af demagnetization.
* Duration of acid leaching (hr). Brackets indicate *in situ* directions.

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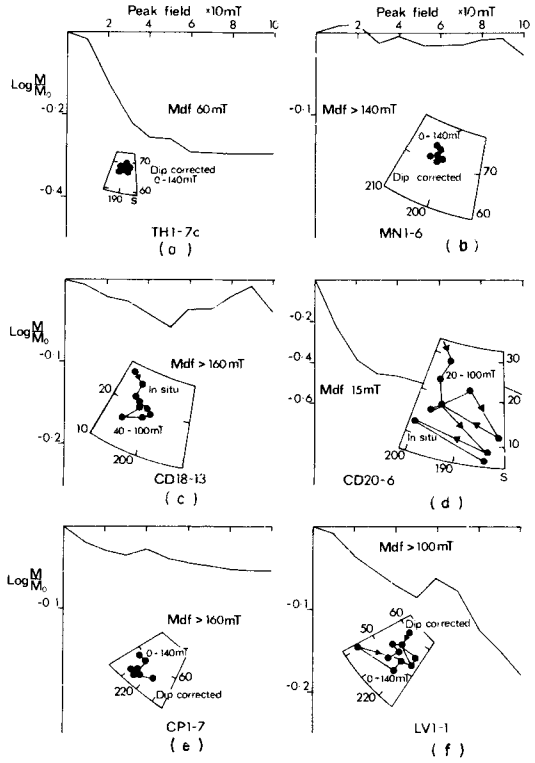


Figure 3. Progressive af demagnatization of NRM in some Armorican redbed sites, with corresponding directional changes indicated (all downward directions).

NRM on leaching are variable between sites. Endpoints are distributed from steep south-west downward directions to shallow south or south-west downward directions (Table 2).

Two specimens from each of the 22 sites stable to af demagnetization were thermally demagnetized in air at progressively higher temperatures with 50° C increments. Many sites show a near linear decay of NRM and blocking temperatures are distributed from 0 to 650°C. Heating at optimum cleaning temperatures suggested by Briden's (1972) stability index (100–200°C) removes a low blocking temperature component A, but fails to separate two or more intermediate and high blocking temperature components of the remaining NRM.

1.4 ANALYSIS OF MULTICOMPONENT NRM

The thermal, acid and af demagnetization trends suggest that two groups of redbed sites with different types of NRM can be distinguished by the different directions of the medium and high blocking temperature (or less soluble or higher remanent-coercivity) components of the NRM.

Type 1 NRM is recognized in all sites except those from the Plourivo-Bréhec succession and Normandy sites MN3 and LN4, and consists generally of three components A, B3 and C. Component A has low remanent-coercivities (< 20 mT), blocking temperatures less than 150°C and scattered steep directions. Component B3 has intermediate blocking temperatures of 150–400 or 500°C (Fig. 5 (a)–(c)). The fraction carrying component B3 is acid soluble in some sites (e.g. Fig. 5(g)) but not in others (Fig. 5(h)). Component C has the highest

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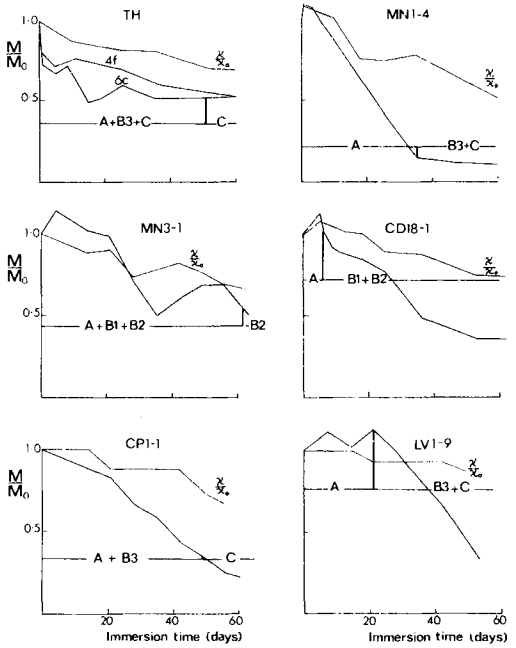


Figure 4. Demagnetization of the NRM in some Armorican redbed sites by acid leaching in 10N HCl. The normalized decay of susceptibility and the components removed by varying amounts of leaching are also shown. In Fig. 4a the susceptbility curve refers to specimen 4f.

blocking temperature of up to about 600 or 650°C. The critical blocking temperature above which only C is recorded and below which B3 and C have overlapping spectra varies between and within type 1 sites from about 400 to 500°C. The magnetic fraction carrying component C is resistant to acid leaching.

Demagnetization trends for type 1 sites are characteristically V shaped (e.g. Fig. 5(a) and (b)). On thermal demagnetization the resultant NRM vector rotates toward the resultant NRM direction (**B3 + C**) of the two higher blocking temperature components in a plane containing this direction and the direction of the total NRM (**A + B3 + C**). Further demagnetization of component B3 causes the resultant vector to rotate in the plane containing B3 and C directions toward C (e.g. Fig. 5(a) and (b)). Component B3 was estimated by the vector subtraction (**B3 + C**) − **C** for specimens in which C could be determined as an endpoint (Tables 2 and 3). Although the presence of component C in the NRM of sites UN1, UN2, CF, SR1 and SR3 was indicated by planar rotation of the resultant NRM vector during demagnetization, it was not possible to isolate component C as an endpoint in these sites owing to its very low intensity relative to noise, nor to estimate component B3 because of overlapping spectra.

Specimens with type 2 NRM are derived from the Plourivo-Bréhec succession, and Normandy sites MN3 and LN4. Their NRM comprises three components, A as in type 1 sites, and intermediate to high (> 150°C) blocking temperature components B1 and B2. Components B1 and B2 have overlapping spectra, but B2 is distinguished as having generally higher blocking temperatures than B1 and a different southerly declination, although both B1 and B2 are shallowly inclined south to south-west. In some sites B2 is the more easterly component, but in others B1 has a more easterly declination than B2. The magnetic fractions carrying components B1 and B2 are equally prone to leaching by acid, and this results in large oscillations in declination between B1 and B2 directions on acid demagnetization

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Table 3. Thermal demagnetization of type 1 NRM in the Armorican redbeds.

Site	Specimen	Resultant B3 + C			Component C			Component B3		
		M	*D*	*I**	*M*	*D*	*I**	*M*	*D*	*I**
TH	1–2D	4.1	204	53	2.1	274	80	2.6	196	27
	1–9C	4.7	205	62	2.2	209	79	2.6	204	48
	1–20B	4.0	192	51	1.9	221	64	2.9	179	37
MN1	1–5	7.7	197	20	4.8	237	50	5.4	175	−10
MN2	2–1	3.0	205	52	0.7	221	58	2.3	206	50
	2–4	B3 endpoint at $D = 210$, $I = 53$								
CD10	10–1	0.6	202	29	0.2	231	63	0.4	215	41
CD26	26–2	4.1	184	36	2.6	113	17‡	3.8	227	26
CP	1–11	6.4	218	48	2.8	216	57	3.7	219	41
	1–12	6.7	229	44	2.3	226	57	4.5	230	37
CR	4	B3 endpoint at $D = 234$, $I = 39$								
LV	1–3	3.8	229	31	1.8	217	73	2.7	(227 231	9)† 4
	1–7	3.8	231	29	2.0	218	71	2.5	(228 231	5)† −1
SR2	2–4	3.8	223	50	1.6	242	56	3.1	206	43
	2–7	2.8	202	53	2.4	233	65	0.9	(197 176	−7)† 5

Notes:

• Dip corrected directions unless bracketed (*in situ*).
† Specimens with shallow B3 directions comparable to B1 or B2 directions (type 2 NRM).
‡ High blocking temperature NRM is multi-component. Notation as in Table 2.

Table 4. Thermal demagnetization of type 2 NRM in the Armorican redbeds.

Site	Specimen	Resultant B1 + B2			Component B2			Component B1		
		M	*D*	*I**	*M*	*D*	*I**	*M*	*D*	*I**
MN3	3–4	3.1	214 231	16 −37	1.7	211 210	34 −32	1.6	217 257	− 4 −38
	3–6	3.5	214	6	0.7	190 221	12 −60	2.9	219 238	4 −41
LN4	4–3	0.4	208 205	−16 − 4	0.3	217 215	−11 − 1	0.1	178 181	12 30
CD18	18–3	5.2	187 184	17 − 6	2.6	174 168	27 − 4	2.7	206 199	33 − 7
	18–9	4.0	194 193	12 − 6	1.4	188 183	19 − 4	2.7	200 198	20 − 7
CD20	20–2	5.9	198 196	5 −10	2.9	172 178	22 12	3.7	216 214	− 9 − 1
	20–3	6.4	210 209	6 −12	2.7	162 163	6 1	5.0	233 233	4 7
CD22	22–11	5.7	208 208	− 3 6	3.4	214 214	− 3 7	2.3	199 199	− 3 −17
	22–13	3.7	201 201	− 4 4	0.9	242 242	6 16	3.0	189 189	− 7 −10

Notes:

• *In situ* directions are shown above dip corrected directions; both are given because it is not possible to determine whether B1 and B2 were acquired before or after folding (see text and Fig. 6). Notation as in Table 2.

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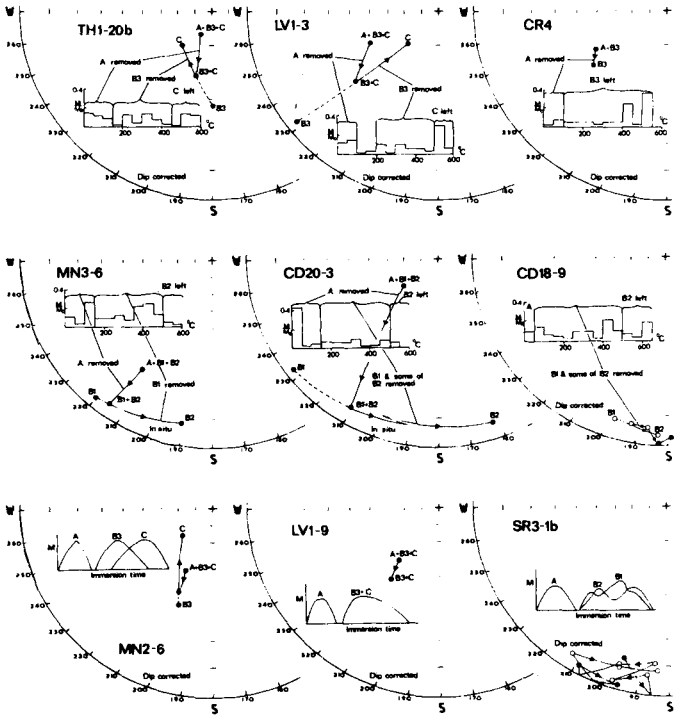


Figure 5. (a)–(f) Thermal demagnetization characteristics of type 1 (a)–(c) and type 2 (d)–(f) multicomponent NRM in the Armorican redbeds, showing the correspondence between blocking temperature spectra and directional changes for each component. (g)–(i) Characteristic directional NRM changes during acid leaching of one (h) or more (g) of components A and B3 (or B1 and B2: i). The relative 'solubility' spectra of each component are shown schematically. Closed symbols: downward directions; open symbols: upward directions (stereographic projections).

(e.g. Fig. 5(i)). Similar declination swings occur during thermal demagnetization of some specimens (e.g. Fig. 5(e) and (f)). The direction of component B2 was determined from the endpoints of demagnetization, and the direction of component B1 by the vector subtraction **(B1 + B2) – B2** (Table 4).

1.5 ORIGINS OF THE VARIOUS COMPONENTS

1.5.1 Component A

The theoretical blocking temperature curves for hematite (Pullaiah *et al.* 1975) suggest that magnetizations with blocking temperatures less than 150°C can be unblocked and remagnetized in less than a day at room temperature. Component A is probably therefore of recent viscous origin, and this is consistent with a storage test which showed that many specimens changed their direction and intensity of total NRM over a period of a few months.

1.5.2 Components B3, B1 and B2

The maximum blocking temperatures of component B3 (in sites with type 1 NRM) of 400–500°C reflect the extent to which component C with higher blocking temperatures

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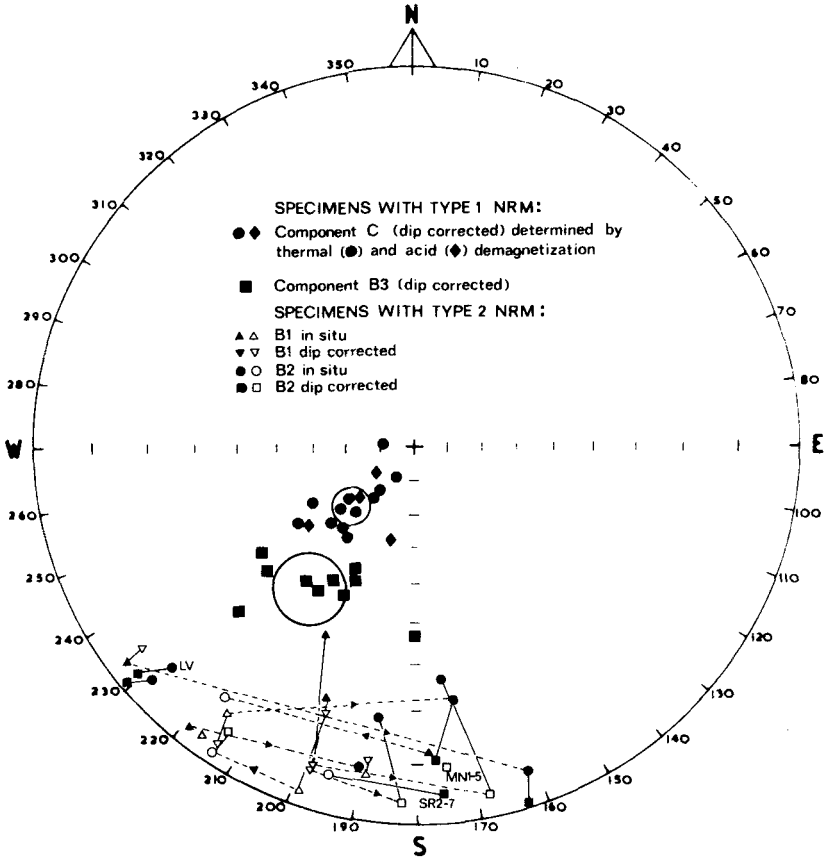


Figure 6. Directions of components B3 and C from specimens with type 1 NRM, and of components B1 and B2 from specimens with type 2 NRM. The circles of 95 per cent confidence for the overall means of seven site mean directions are shown for B3 and C. Dashed tie lines connect B1 and B2 directions from a single specimen; the arrow is toward B2. Closed symbols: downward directions; open symbols: upward directions (stereographic projections).

has been overprinted by component B3 (Fig. 5). The blocking curves for hematite (Pullaiah *et al.* 1975) imply that temperatures of about 200°C (burial metamorphic facies) over a prolonged period (> 1 Myr) would have been sufficient to partially overprint component C in type 1 sites, and for B3 to have been acquired as a viscous PTRM. Component B3 was probably therefore stabilized on uplift after acquisition during burial. In some specimens, B3 occupies the entire blocking temperature range above about 150°C (e.g. Fig. 5(c)), and this may reflect complete overprinting of C. Comparison of the precisions of *in situ* and dip-corrected groupings of B3 directions determined from endpoints and vector analysis implies with 95 per cent confidence (Cox 1969) that component B3 was acquired in most sites before Westphalian folding of the redbeds. The dip-corrected mean direction (Fig. 6) of B3 for seven sites (10 specimens) is $D = 216°$, $I = 41°$ ($\alpha_{95} = 10°$), which yields a palaeomagnetic pole at 324°E 11°S (Table 5). Comparison with the European apparent polar wander (apw) path (Fig. 7) implies that component B3 was acquired during the Devonian or early Carboniferous, perhaps during the uplift associated with the Bretonic phase of the Hercynian Orogeny.

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Table 5. Mean directions of NRM and palaeomagnetic pole positions of components B3 and C in the Armorican redbeds, the Erquy Spilite Series and the Trégastel–Ploumanac'h granite.

Component/rock unit	*N*	*D*	*I*	*R*	*k*	a_{95}	Long.	Lat.	dp	dm
Redbed component B3	7 sites	216	41	6.83	35.6	10	324 E	11 S	8	12
(dip-corrected)	10 spec.	214	43	9.72	31.7	9	326 E	11 S	7	11
Redbed component C	7 sites	228	63	6.91	70.6	7	325 E	11 N	9	11
(dip-corrected)	16 spec.	226	65	15.71	52.4	5	328 E	13 N	7	8
Erquy Spilite Series (not dip-corrected)	5 sites	221	82	4.92	47.2	11	344 E	35 N	21	22
Trégastel–Ploumanac'h granite complex	5 sites	200	9	4.97	121	7	332 E	34 S	4	7

Abbreviations:

D, *I* declination, inclination of NRM; *R* resultant of *N* unit vectors; *k* estimate of Fisher's (1953) precision parameter; a_{95} semi-angle of the cone of 95 per cent confidence about the mean direction; dp, dm semi-angle subtended by major and minor axes of ellipse of 95 per cent confidence about the mean pole position. Analysis of directions follows Fisher (1953).

In specimens with type 2 NRM it is not possible to determine whether components B1 and B2 were acquired after or before folding, because dip-corrected directions are as scattered as *in situ* directions (Fig. 6, Table 4). This may be partly because B1 and B2 were acquired at about the same time as the late Carboniferous folding; the inclinations of B1 and B2 are

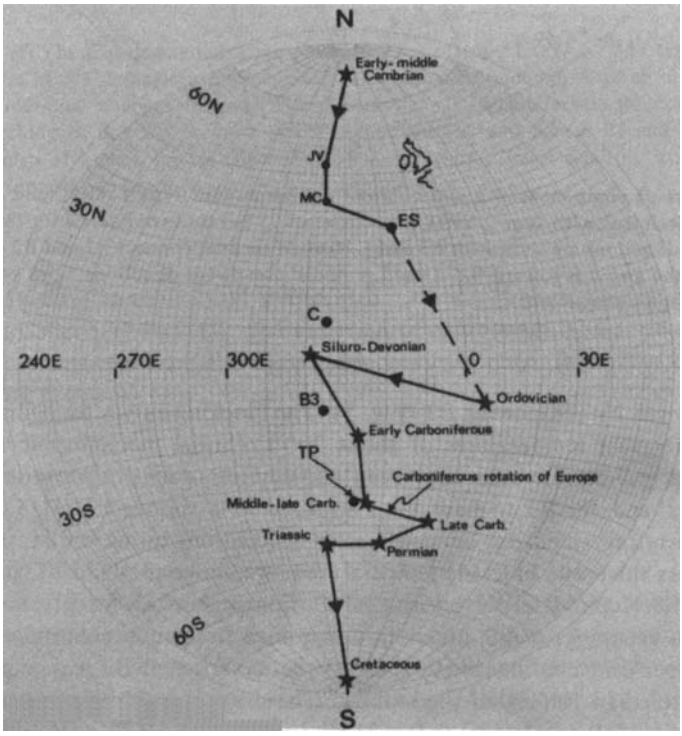


Figure 7. Palaeomagnetic pole positions corresponding to the Armorican redbed components (B3 and C), and for the Erquy Spilite Series (ES) and Trégastel–Ploumanac'h complex (TP) in relation to the Phanerozoic apparent polar wander path for Europe suggested by Duff (1978b). The path segment joining pole ES with the mean Ordovician pole is tentative. JV: pole for the Jersey Volcanics (533 Myr); MC: Mona Complex pole (*in situ*). Equatorial equal area projection.

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very similar to those in rock units such as the Trégastel-Ploumanac'h granite with middle to late Carboniferous NRM. However, the *in situ* and dip-corrected directions of B1 and B2 in some type 2 sites are similar because of the easterly or westerly shallow dipping nature of the redbeds, and this suggests that the scatter, which is mainly one of declination, originated by some process other than folding about shallowly inclined meridional axes. Possible origins of this scatter are:

(1) Preferred orientation of DRM in the bedding plane, with an azimuthal scatter of directions. However, it is likely that components B1 and B2 were acquired during demagnetization. In addition, the SSW declinations are inconsistent with the north-west to north-east declinations in Cambrian and Ordovician igneous rocks in the Armorican Massif (Duff 1978b).

(2) Rotation of the direction of NRM toward a horizontal, oblate magnetic fabric. The homogeneous, finely divided nature of the hematite probably precludes large magnetic anisotropy.

(3) Failure to separate components B1 and B2 owing, perhaps, to overlapping blocking temperature (or 'solubility') spectra, so that the estimate of NRM directions lies in a plane between these components. This probably only accounts for the declination scatter in a few specimens which show declination swings during demagnetization.

(4) Local tectonic rotations in the Armorican Massif associated with the Hercynian Orogeny, or magnetization during coherent late Carboniferous rotation of the bulk of Europe about a vertical axis. Although variable local rotations would best explain why in some specimens B2 is east of B1, while in others it is the more westerly component, there is little geological evidence for differential rotations of the Côtes-du-Nord redbeds about vertical axes. Coherent rotations about a Euler axis in or near Europe however, is consistent with results from the Bohemian Massif (Birkenmajer, Krs & Nairn 1968) and with Carboniferous palaeomagnetic pole positions from elsewhere in Europe (Duff 1978b). It suggests that if components B1 and B2 have been correctly separated, they must have been acquired at least partly during rotation to explain the great variability in their declination difference (either *in situ* or dip-corrected). Coherent rotation could be consistent with the variable easterly or westerly declination of B2 relative to B1 if these components were of variable relative age.

1.5.3 Component C

Comparison of the maximum blocking temperatures of component C (about 600–650°C) with the theoretical blocking curves for hematite (Pullaiah *et al.* 1975) implies that the redbeds would have had to have been heated to greater than 500°C for a sustained period (> 1 Myr) for C to have been acquired as a viscous PTRM. Since there is no evidence of such metamorphism, component C was probably acquired as a CRM rather than a viscous PTRM. Comparison of *in situ* and dip-corrected C directions determined from the end-points of thermal and acid demagnetization implies with 95 per cent confidence (Cox 1969) that component C was acquired before Westphalian folding. The dip-corrected mean direction of C from seven sites, $D = 228°$, $I = 63°$ (16 specimens; Fig. 6), yields a palaeomagnetic pole at 325° E 11° N (Table 5, Fig. 7). Comparison with the European apw path implies that component C was acquired during the Silurian or early Devonian.

1.6 CARRIERS OF NRM IN THE RED SHALE OF THE ROZEL CONGLOMERATE

To obtain further insight into the origin of the intermediate and high blocking temperature components B3 and C, and their possible carriers, the nature of the Fe–Ti oxides and the

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magnetic properties of the red shale at the base of the Rozel Conglomerate (Jersey) were investigated.

1.6.1 Magnetic properties

The remanent-coercivity spectra (Dunlop 1972) for a specimen leached in 10N HCl and an unleached specimen from the same sample were derived from hysteresis of IRM curves, and are shown below the initial magnetization curves in Fig. 8. Both leached and unleached specimens failed to saturate in fields of 1000 mT. There is a marked difference, however, in the coercivities of unsaturated remanence (H_R); for the unleached and leached specimens H_R is 500 mT and 320 mT respectively. The remanent-coercivity spectra are also quite distinctive. According to Dunlop's (1972) criteria, the spectrum of the unleached specimen implies that the IRM is carried mainly by hematite pigment and some specularite. The remanent-coercivity spectrum is displaced after leaching of the red pigment toward lower values, so that the bulk of IRM resides in the range 0–600 mT (Fig. 8). This implies that the harder IRM in the red shales is carried by the red pigment leached out by acid, and that residual specularite probably carries the bulk of IRM in the leached specimen.

These results are consistent with high field thermomagnetic analyses and imply that NRM in the red shale is carried by pure hematite, which is present both as pigment and specularite. Although the pigment is magnetically harder that the specularite, thermal demagnetization in conjunction with acid leaching suggests that most of the pigment has lower blocking temperatures than the specularite. Concurrence of endpoints with the component C direction after both acid and thermal demagnetization, and the similar leaching and thermal demagnetization trends imply that components A and B3 are carried by soluble hematite pigment, and that the high blocking temperature component C is mainly carried by the insoluble specularite.

1.6.2 Origin and composition of the hematite

Hematite is present in the red shales as part of the interstitial cement which binds clasts, and as discrete specularite grains independent of the pigment. In addition, most iron silicate

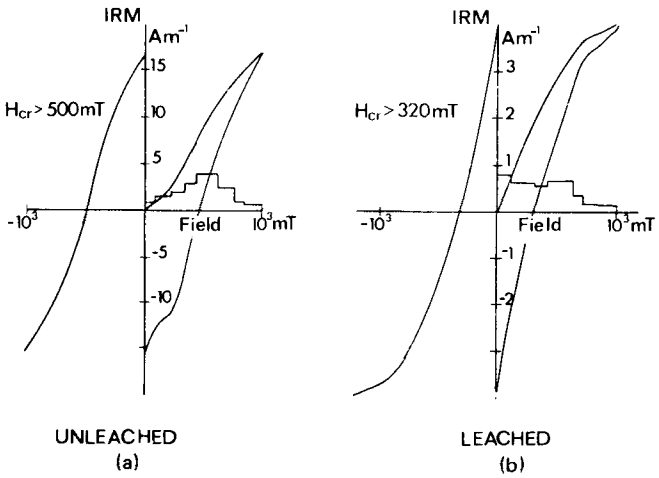


Figure 8. IRM hysteresis curves for unleached (a) and leached (b) specimens of the Rozel red shale. The remanent-coercivity spectra correspond to the initial magnetization curves. H_{cr} is the coercivity of saturation remanence.

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clasts are partially replaced by hematite pigment. Biotite is preferentially replaced along its cleavage, and relict amphiboles and volcanic rock fragments have distinct hematite haloes. Only a few grains of relict detrital magnetite are observed.

These textures imply that the interstitial pigment may have been derived from the post-depositional decomposition of former iron silicate grains during authigenesis. This is consistent with the results of Walker (1967, 1974), who has shown that the formation of red pigment in redbeds occurs during diagenesis when iron bearing detrital grains alter first to iron hydroxides then to authigenic hematite, depending on the local interstitial Eh/Ph environment. He cites the presence of haloes around detrital grains, as in the red shales of the Rozel Conglomerate, as diagnostic of this process. The specularite may also have crystallized during post-depositional processes (Turner & Ixer 1977). Acquisition of CRM associated with authigenic growth of the pigment and specularite from detrital grains is consistent with the absence of a Cambro-Ordovician DRM in the redbed sites.

1.7 MAGNETIZATION HISTORY OF THE ARMORICAN REDBEDS

The palaeomagnetic results from the Armorican redbeds and the more detailed studies of the red shale from the Rozel Conglomerate suggests the following history:

(1) Clastic deposition of iron silicates, magnetites and possibly specularite during the Cambrian or early Ordovician.

(2) Acquisition during Silurian or early Devonian times of component C, the oldest magnetization, probably as a CRM associated with the growth of specularite and possibly less soluble hematite pigment.

(3) Acquisition of component B3 during remagnetization of most sites other than those in the Plourivo-Bréhec succession in Devonian to early Carboniferous times. Component B3 was probably acquired as a viscous PTRM during burial, which variably overprinted the older component C and was stabilized on uplift – perhaps during the Bretonic phase of the Hercynian Orogeny.

(4) Acquisition in the red successions of Plourivo-Bréhec and some other sites (MN3 and LN4) of components of B1 and B2, possibly also as viscous PTRMs, during the late Carboniferous. These have overprinted any record of older (C, B3 or other) magnetizations. The most likely origin of the declination scatter is magnetization before, after and possibly during the late Carboniferous rotation of Europe.

2 The Erquy Spilite Series

2.1 INTRODUCTION

The Erquy Spilite Series (Auvray 1968) crop out on the north-eastern side of the Baie de Saint-Brieuc on the north coast of Brittany. Collections for palaeomagnetic study were made by *in situ* drilling at five sites distributed through the well-exposed section on Pointe de la Heussaye, immediately south-west of Erquy (Fig. 9). Auvray (1968) has distinguished pillow lavas, basic sills, breccias, tuffs and sediments in this section. The controversial Lower Ordovician age assigned to the Erquy Spilite Series by Vidal *et al.* (1971) and Deunff *et al.* (1973) was based on microfauna and Rb-Sr whole rock results obtained from lutites, pillow lavas and keratophyres in the section.

The asymmetry of metabasaltic pillows indicates that the entire section is overturned; it dips very steeply southward, but 'youngs' northward. Two sites, CD3 and CD4, are located in pillow metabasalt toward the northern extremity of Pointe de la Heussaye. Sites CD5,

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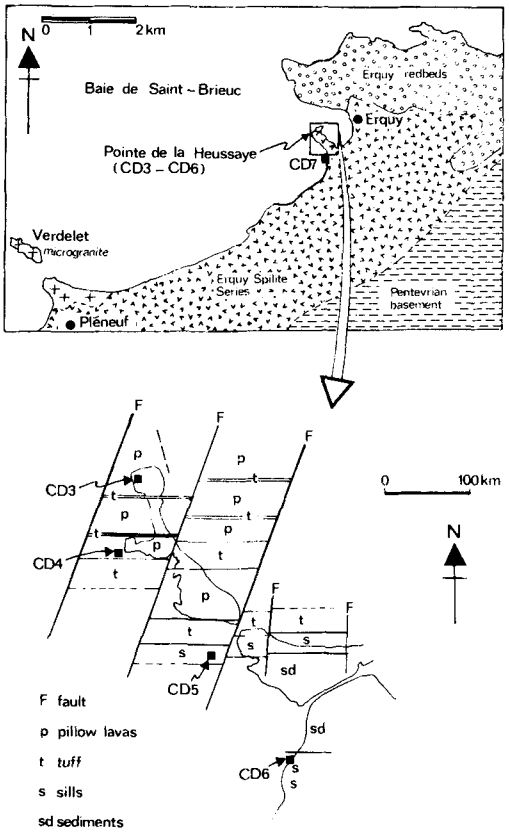


Figure 9. Geological map of the Erquy Spilite Series showing the distribution of palaeomagnetic sites in the succession of Pointe de la Heussaye north of Pléneuf (map after Auvray & Hameurt, 1971).

CD6 and CD7 are located in fine-grained dolerite sills which are intercalated in the metabasalts, tuffs and sediments in different parts of the section toward the mainland (southeast) end of Pointe de la Heussaye. Petrographic and geochemical studies suggest that the sills are cogenetic with the metabasalts (Auvray 1968) and can therefore be regarded as part of the Erquy Spilite Series. The dip of the sills was estimated from adjacent tuffs or sediments.

2.2 NRM AND AF DEMAGNETIZATION

Intensities of NRM in the metabasalts and sills are typically of the order 10^{-1}–1 A/m. The five site mean directions of total NRM (*in situ*) are distributed about a steep overall mean direction $D = 264°$, $I = 81°$ ($\alpha_{95} = 21°$). After correction for structural dip by restoring the sequence so that it 'youngs' upward, the five site mean directions are distributed about $D = 347°$, $I = -5°$ ($\alpha_{95} = 19°$; Fig. 10). Both *in situ* and dip-corrected overall mean directions of total NRM are oblique and significantly different at the 95 per cent probability level from the direction of the present field at Erquy ($D = 352°$, $I = 65°$).

The NRM of at least two specimens from each site was progressively af demagnetized in peak fields up to 100 mT, and an example of the corresponding change in intensity and direction of NRM of a specimen from four sites is shown in Fig. 11. Half of the initial NRM is demagnetized in fields of 0–10 mT. There is a marked decrease in the decay of NRM in

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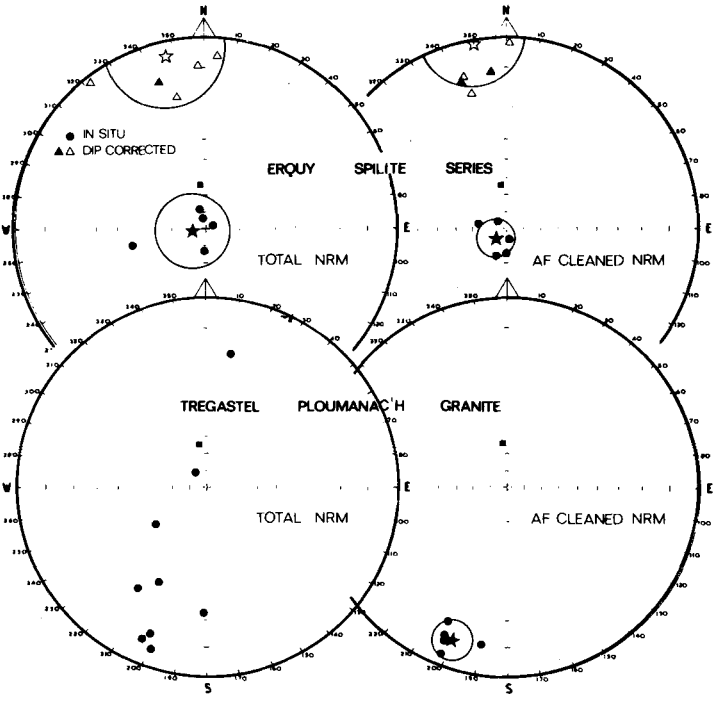


Figure 10. Site and group (stars) mean directions of NRM for the Erquy Spilite Series (both *in situ* and dip-corrected) and the Trégastel–Ploumanac'h granite before and after optimum af cleaning. Group mean directions are shown with their circles of 95 per cent confidence. Closed symbols: downward directions; open symbols: upward directions; present field direction indicated by square (stereographic projections).

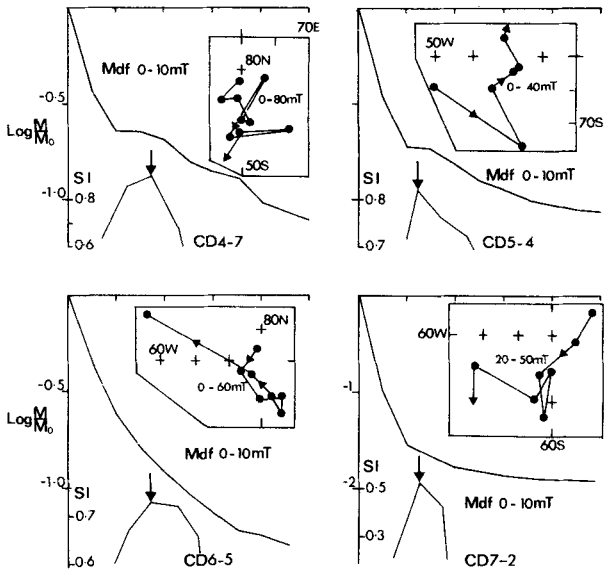


Figure 11. Progressive af demagnetization of the NRM in sites from the Erquy Spilite Series. Corresponding directional changes (all directions *in situ* and downward) and variation of stability indices with peak field are also shown. Mdf is the median destructive field.

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specimens from the metabasalt of CD4 and the sill of CD5 over the range of peak fields from 20 to 40 mT, which reflects the separation of a soft component with low coercivities of remanence below 20 mT from a harder component of NRM. The NRM of specimens from the other sites decayed continuously with increasing peak fields. The optimum peak field for site cleaning, corresponding to the maximum stability indices (indicated by arrows in Fig. 11) varied between sites according to the variation in the remanent-coercivity spectra of the harder component of NRM.

2.3 ANALYSIS OF NRM DIRECTIONS

After demagnetization at the optimum peak fields *in situ* site mean directions of NRM converged toward an overall mean direction $D = 221°$, $I = 82°$ ($\alpha_{95} = 11°$; Fig. 10, Table 5). Dip-corrected site mean directions (Fig. 10) yielded an overall mean direction $D = 350°$, $I = -1°$ ($\alpha_{95} = 15°$). A two-tier analysis of stable NRM directions (Watson & Irving 1957) shows that the between-site scatter is significant at the 95 per cent probability level for both *in situ* and dip-corrected results ($F_{8,50} = 8.5$ and 17.3 respectively), and that it is larger for the dip-corrected results.

Although the estimate of precision for the overall mean direction determined *in situ* ($\beta = 55$) is nearly twice that for the dip-corrected overall mean ($\beta = 28$), a test of the relative age of NRM and folding (Cox 1969) is inconclusive; there is insufficient variability in the attitude of bedding between the five sites. However, for the following reasons a stable magnetization in the Erquy Spilites is considered to have been most likely acquired after folding:

(1) The Erquy Spilites were folded in the Late Precambrian or early Cambrian during the Cadomian Orogeny. This is indicated by unconformable relationships with the Cambrian redbeds and arkoses of Erquy and Cap Fréhel.

(2) A metamorphic event overprinting the primary paragenesis occurred after this folding, and was associated with isotopic rehomogenization of the Rb-Sr system during the early Ordovician (Brown & Roach 1972).

It is very probably that stable NRM in the Erquy Spilites was acquired – perhaps as a PTRM – during this post-folding event, and this interpretation is consistent with a middle to late Cambrian age determined for the *in situ* magnetization by comparison of the corresponding pole with the polar wander path for Europe (Duff 1978b). The palaeomagnetically significant mean direction of NRM for the Erquy Spilites is therefore determined *in situ* as $D = 221°$, $I = 82°$, which corresponds to a palaeomagnetic pole position at 344° E 35° N (Table 5).

3 The Trégastel-Ploumanac'h granite complex

3.1 INTRODUCTION

The Trégastel-Ploumanac'h granite is a high level, subvolcanic complex on the north coast of Brittany west of Perros-Guirec (Côtes-du-Nord). It consists of three main granite phases (Fig. 12) that were emplaced successively, so that the youngest phase forms the core of the complex (Barrière 1976). Features typical of high level granites such as cone sheets, ring dykes and autobreccias are present in the outer granite phase. An Rb-Sr whole rock age of 300 ± 5 Myr (middle Carboniferous) and an incidental K-Ar biotite age (300 ± 6) indicates that the complex was emplaced and cooled very rapidly (Adams 1976). Stable NRM is therefore the most likely to have been acquired about 300 Myr ago. Nine sites were

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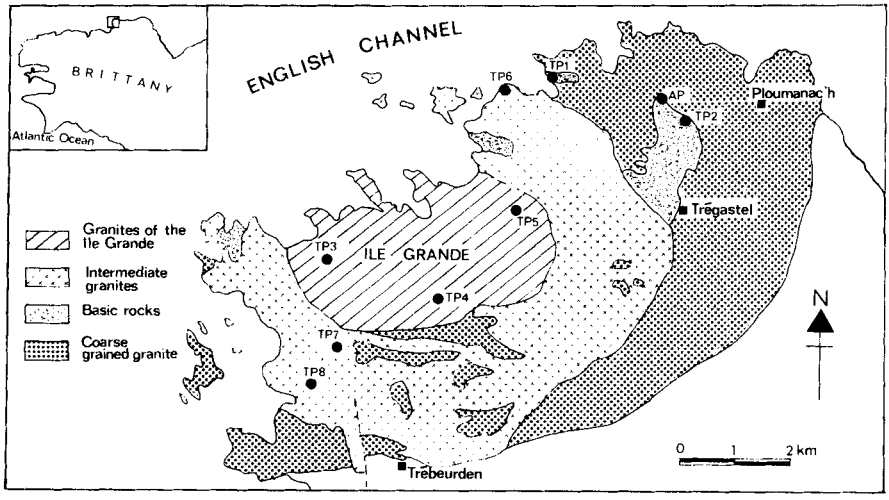


Figure 12. Geological map (after Barrière 1976) of the Trégastel–Ploumanac'h granite complex of northern Brittany, showing the location of palaeomagnetic sites.

sampled in the Trégastel-Ploumanac'h complex: three in the diorites; three in the 'intermediate' granites, and three in the youngest phase, the granite of the Ile Grande (Fig. 12). The outermost granite phase, which intrudes gneissic basement, was too coarse for palaeomagnetic study.

3.2 ANALYSIS OF NRM AND PALAEOMAGNETIC POLE POSITION

The nine site mean directions of total NRM (Fig. 10) are distributed close to a SSE trending plane which contains the present field direction. This implies the presence of soft VRM

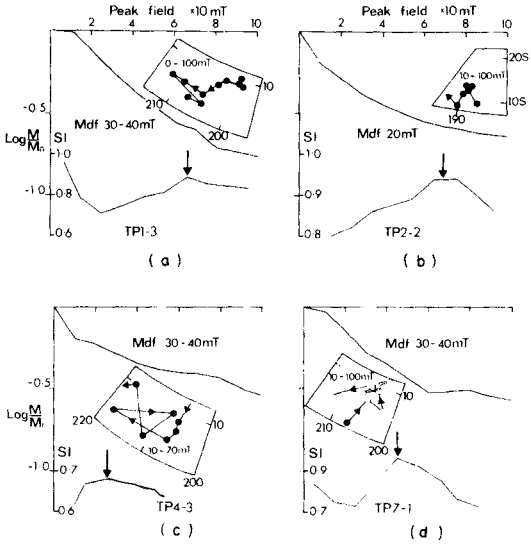


Figure 13. Progressive af demagnetization of the NRM in diorites and granites of the Trégastel–Ploumanac'h complex. Corresponding directional changes (closed symbols: downward directions; open symbols: upward directions) and variation of stability indices with peak field are also shown.

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components. Intensities of NRM vary from about 10^{-4} to 10^{-2} A/m for the fine-grained granite sites, and from about 10^{2} to 10^{-1} A/m for the diorites.

Only five of the nine sites were stable against af demagnetization. The unstable sites were all weakly magnetized fine-grained granites, which showed large erratic changes of NRM direction during demagnetization at low peak fields. The most stable rocks were the diorites, which showed great consistency of NRM directions in peak fields up to at least 100 mT, and considerable resistance to demagnetization (Fig. 13(a) and (b)). Sites TP4 and TP7 from the Ile Grande and 'intermediate' granites respectively were the only granite sites stable against af demagnetization. The median destructive fields (Mdf) for these granites were similar to those for the diorites (20–40 mT), although a different range of remanent-coercivities was indicated by a narrower range of high stability indices with maxima at peak fields of 30–40 mT (Fig. 13(c) and (d)).

After optimum field demagnetization the within-site precision of four of the stable sites was greatly increased. The overall mean direction of stable NRM for the five sites is $D = 200°$, $I = 9°$ ($\alpha_{95} = 7°$; Table 5), which yields a palaeomagnetic pole at 332° E 34° S ($dp = 4°$, $dm = 7°$), close to middle to late Carboniferous poles for Europe.

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